

Lecture 7: Proof Notes for Instructor

Duality Theory

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Abstract

These notes provide detailed proofs and teaching guidance for the main results in duality theory: weak duality, strong duality via Slater's condition, complementary slackness, and applications to fractional knapsack problems. The emphasis is on intuition and connections rather than memorization.

1 Background and Setup

Throughout these notes, we consider optimization problems of the form:

$$\min_x f(x) \tag{1}$$

$$\text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, m \tag{2}$$

$$h_j(x) = 0, \quad j = 1, \dots, p \tag{3}$$

where $f, g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition 1 (Lagrangian). The **Lagrangian** is defined as:

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

where $\lambda \in \mathbb{R}^m$ and $\nu \in \mathbb{R}^p$ are the **dual variables** (also called Lagrange multipliers).

Definition 2 (Dual Function). The **dual function** (or Lagrangian dual) is:

$$g(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$$

The domain of g is the set of (λ, ν) for which $g(\lambda, \nu) > -\infty$.

Definition 3 (Dual Problem). The **dual problem** is:

$$\max_{\lambda, \nu} g(\lambda, \nu) \quad \text{subject to } \lambda \geq 0$$

We denote $p^* = \inf_x f(x)$ subject to constraints (primal optimal) and $d^* = \sup_{\lambda, \nu: \lambda \geq 0} g(\lambda, \nu)$ (dual optimal).

2 Theorem 1: Weak Duality

Theorem 1 (Weak Duality). *For any $\mathbf{x}$ feasible to the primal problem and any (λ, ν) with $\lambda \geq 0$, we have:*

$$g(\lambda, \nu) \leq f(\mathbf{x})$$

Therefore, $d^* \leq p^*$.

2.1 Complete Proof

Proof. Suppose $\mathbf{x}$ is feasible to the primal, meaning:

- $g_i(\mathbf{x}) \leq 0$ for all $i = 1, \dots, m$
- $h_j(\mathbf{x}) = 0$ for all $j = 1, \dots, p$

Let (λ, ν) be arbitrary with $\lambda \geq 0$. By definition of the dual function:

$$g(\lambda, \nu) = \inf_z \mathcal{L}(z, \lambda, \nu) \leq \mathcal{L}(\mathbf{x}, \lambda, \nu)$$

Expanding the Lagrangian at $\mathbf{x}$:

$$\mathcal{L}(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x})$$

Since $h_j(\mathbf{x}) = 0$ for all j , the last sum vanishes:

$$\mathcal{L}(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x})$$

Now observe that each term in the sum satisfies:

- $g_i(\mathbf{x}) \leq 0$ (feasibility of $\mathbf{x}$)
- $\lambda_i \geq 0$ (dual feasibility)

Therefore, $\lambda_i g_i(\mathbf{x}) \leq 0$ for each i , which gives:

$$\sum_{i=1}^m \lambda_i g_i(\mathbf{x}) \leq 0$$

Combining these results:

$$g(\lambda, \nu) \leq \mathcal{L}(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) \leq f(\mathbf{x})$$

This holds for any feasible $\mathbf{x}$ and any (λ, ν) with $\lambda \geq 0$. Taking the supremum over feasible dual variables:

$$d^* = \sup_{\lambda, \nu: \lambda \geq 0} g(\lambda, \nu) \leq f(\mathbf{x})$$

Taking the infimum over all feasible primal points:

$$d^* \leq \inf_{\text{primal feasible } \mathbf{x}} f(\mathbf{x}) = p^*$$

□

2.2 Teaching Notes

Key Insight. What Makes Weak Duality Powerful: *This result requires no convexity assumptions. It holds for any problem with any structure. The only requirement is feasibility. This is a fundamental inequality that always gives us a lower bound on the optimal primal value.*

Remark (Key Points for Students). 1. **The role of constraints:** *The inequality $\lambda_i g_i(\mathbf{x}) \leq 0$ is the entire reason weak duality works. It combines the constraint satisfaction ($g_i(\mathbf{x}) \leq 0$) with the dual variables being non-negative.*

2. **Why equality constraints have no sign restriction:** *The dual variables ν_j for equality constraints can be any sign, because $\nu_j h_j(\mathbf{x}) = 0$ regardless. This is why ν is unrestricted.*

3. **The duality gap:** *We call $p^* - d^*$ the **duality gap**. Weak duality guarantees this is non-negative. When the gap is zero, we have **strong duality**.*

4. **Non-convex problems:** *Students are often surprised that weak duality holds even for non-convex problems. Emphasize: the dual function $g(\lambda, \nu)$ is always concave (it's a pointwise minimum of affine functions), even if the primal is non-convex. This is why the dual problem is always convex.*

Remark (Common Misconception). *Students sometimes think weak duality should be “tight” (i.e., an equality). Clarify that the inequality can be strict, and when it is, the duality gap is positive. This motivates Slater’s condition and strong duality.*

Remark (Board Diagram Suggestion). *Draw a two-variable example: a simple non-convex feasible region (e.g., a non-convex set in 2D) and illustrate:*

- A feasible point $\mathbf{x}$ at level $f(\mathbf{x}) = 10$
- The dual lower bound as a horizontal line at level, say, $d^* = 8$
- Show $f(\mathbf{x}) \geq d^*$ geometrically

This makes the weak duality inequality concrete.

3 Theorem 2: Strong Duality via Slater's Condition

Definition 4 (Slater's Condition). *For a convex problem, **Slater's condition** holds if there exists a point $\tilde{x}$ such that:*

- $g_i(\tilde{x}) < 0$ for all $i = 1, \dots, m$ (strict inequality)
- $h_j(\tilde{x}) = 0$ for all $j = 1, \dots, p$ (equality constraints must still be satisfied)
- $\tilde{x} \in \text{relint}(\text{dom}(f))$ (relative interior of domain of f)

In the case of affine equality constraints, just needing $g_i(\tilde{x}) < 0$ for all i suffices.

Theorem 2 (Strong Duality under Slater's Condition). *Consider a convex optimization problem where the objective f and constraint functions g_i are convex, h_j are affine, and $p^* < \infty$. If Slater's condition holds, then:*

$$d^* = p^*$$

and the maximum in the dual problem is attained (if $d^ < \infty$).*

3.1 Proof Sketch

Intuition and Key Steps. The complete proof of strong duality is technical and uses the separating hyperplane theorem. We outline the main ideas:

Step 1: Define the epigraph set

Consider the set:

$$G = \{(u, v, t) : \exists x \text{ s.t. } g_i(x) \leq u_i \ (i = 1, \dots, m), h_j(x) = v_j \ (j = 1, \dots, p), f(x) \leq t\}$$

This set encodes all possible values of constraints and objective. By convexity of f and g_i , the set G is **convex**.

Step 2: Locate the primal optimum in G

The point $(0, 0, p^*)$ is the “boundary” of G : it's in the closure of G (by definition of p^*) but we can approach it from above. Specifically:

- All points $(0, 0, t)$ with $t > p^*$ are in the interior of G (by feasibility and continuity)
- The point $(0, 0, p^*)$ lies on the boundary

Step 3: Apply the separating hyperplane theorem

Since G is convex and $(0, 0, p^*)$ is on its boundary, there exists a **separating hyperplane**. This hyperplane has normal vector $(\lambda, \nu, \lambda_0)$ such that:

$$\lambda^T u + \nu^T v + \lambda_0 t \geq 0 \quad \text{for all } (u, v, t) \in G$$

Step 4: Use Slater's condition to show $\lambda_i \geq 0$

Slater's condition provides a point $\tilde{x}$ with $g_i(\tilde{x}) < 0$ and $h_j(\tilde{x}) = 0$. For any $t > f(\tilde{x})$, the point $(g(\tilde{x}), 0, t) \in G$ (where $g(\tilde{x})$ denotes the vector).

The hyperplane inequality at this point gives:

$$\lambda^T g(\tilde{x}) + \lambda_0 t \geq 0$$

Since $\lambda_i g_i(\tilde{x}) < 0$ and t can be arbitrarily large, we must have $\lambda_0 \geq 0$. A careful argument (considering what happens when $\lambda_0 = 0$) shows $\lambda_0 > 0$. Dividing by λ_0 and rearranging then implies $\lambda_i \geq 0$ and reveals that:

$$p^* = \frac{1}{\lambda_0} \left(-\inf_x [\lambda^T g(x) + \nu^T h(x) + \lambda_0 f(x)] \right) = g(\lambda/\lambda_0, \nu/\lambda_0)$$

This shows $d^* \geq p^*$. Combined with weak duality ($d^* \leq p^*$), we have $d^* = p^*$. $\square$

3.2 Teaching Notes

Key Insight. The Geometric Intuition: *Imagine all the feasible (u, v, t) values as a convex “cloud.” The point $(0, 0, p^*)$ is the “corner” of this cloud (the point with the smallest objective). Slater’s condition ensures this corner is not degenerate (i.e., not hidden at the boundary of a lower-dimensional face). The separating hyperplane then gives us dual multipliers.*

Remark (Why Slater’s Condition is Not Always Needed). • *Strong duality can hold even without Slater’s condition in some problems (e.g., linear programs by other arguments).*

- *However, Slater’s condition is the **generic sufficient condition** for convex problems.*
- *For problems with only affine constraints, a simpler condition (e.g., feasibility) often suffices.*

Remark (Common Student Confusions). 1. **“Why strict inequality?”** *The strict inequality $g_i(\tilde{x}) < 0$ is essential because it gives us room to perturb the constraints. This is crucial for the separating hyperplane argument to produce $\lambda_i \geq 0$. If we only had $g_i(\tilde{x}) \leq 0$, we could not conclude this.*

2. **“What if Slater fails?”** *Then strong duality may not hold, and there could be a duality gap. Present a simple counterexample: minimize $-\sqrt{-x}$ subject to $x \leq 0$. The primal optimum is 0, but the dual bound is $-\infty$ (no duality gap in this case, but the structure fails).*

3. **“Why convexity?”** *Convexity is needed for two reasons: (1) the set G is convex, and (2) the dual function is automatically concave even for non-convex primal. Without convexity of the primal, the separating hyperplane argument fails to produce the exact relationship.*

Remark (Board Diagram Suggestion). *Draw the set G in 3D (or 2D for a simplified problem):*

- *Draw a convex region representing G*
- *Mark the point $(0, 0, p^*)$ on its boundary*
- *Draw a separating hyperplane (a line in 2D, a plane in 3D)*
- *Show how the normal vector of this hyperplane gives the multipliers $(\lambda, \nu, \lambda_0)$*

This geometric picture is worth a thousand words.

4 Theorem 3: Complementary Slackness from Duality

Theorem 3 (Complementary Slackness Conditions). *Suppose strong duality holds (i.e., $d^* = p^*$) and suppose $\mathbf{x}^*$ is primal optimal and (λ^*, ν^*) is dual optimal. Then:*

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \text{for all } i = 1, \dots, m$$

Equivalently:

$$\lambda_i^* > 0 \implies g_i(\mathbf{x}^*) = 0, \quad \text{and} \quad g_i(\mathbf{x}^*) < 0 \implies \lambda_i^* = 0$$

4.1 Complete Proof

Proof. Since $\mathbf{x}^*$ is primal optimal and (λ^*, ν^*) is dual optimal, and strong duality holds, we have:

$$f(\mathbf{x}^*) = d^* = g(\lambda^*, \nu^*)$$

By definition of the dual function:

$$g(\lambda^*, \nu^*) = \inf_z \mathcal{L}(z, \lambda^*, \nu^*) \leq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*)$$

Expanding:

$$\mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* h_j(\mathbf{x}^*)$$

Since $\mathbf{x}^*$ is primal feasible, $h_j(\mathbf{x}^*) = 0$, so:

$$\mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*)$$

Combining the strong duality equation with the inequality from the definition of the dual function:

$$f(\mathbf{x}^*) = g(\lambda^*, \nu^*) \leq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*)$$

The first and last expressions are equal, so:

$$\sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*) = 0$$

Now, we use the fact that each term in this sum is non-positive:

- $g_i(\mathbf{x}^*) \leq 0$ (primal feasibility)
- $\lambda_i^* \geq 0$ (dual feasibility)

Therefore, $\lambda_i^* g_i(\mathbf{x}^*) \leq 0$ for each i .

Since the sum of non-positive terms equals zero, each term must be zero:

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \text{for all } i = 1, \dots, m$$

□

4.2 Teaching Notes

Key Insight. *The Intuition Behind Complementary Slackness:* If a constraint is not active (slack), i.e., $g_i(\mathbf{x}^*) < 0$, then it doesn't "matter" at the optimum, so the dual variable can be zero. Conversely, if a constraint is tight ($g_i(\mathbf{x}^*) = 0$), then it might affect the solution, so λ_i^* could be non-zero. This is the **economic interpretation**: the multiplier λ_i^* represents the "shadow price" of the constraint, and we only pay a shadow price for binding constraints.

Remark (Practical Uses). 1. **Verification tool:** Given a candidate solution $(\mathbf{x}^*, \lambda^*, \nu^*)$, you can quickly check if it's optimal by verifying complementary slackness and feasibility.

2. **KKT Conditions:** Complementary slackness is part of the Karush-Kuhn-Tucker (KKT) conditions, which are necessary (and with convexity, sufficient) for optimality.

3. **Problem structure:** Often, identifying which constraints should be tight at the optimum gives insight into the solution structure.

Remark (Common Confusions). • **Direction of implication:** Complementary slackness says: if $\lambda_i^* > 0$ then $g_i(\mathbf{x}^*) = 0$, NOT the other way around. A constraint can be tight without the dual variable being positive.

• **Necessity vs. Sufficiency:** Complementary slackness is necessary at optimality (when strong duality holds), but not sufficient by itself. You also need stationarity of the Lagrangian.

• **Equality constraints:** Note that complementary slackness does not apply to equality constraints. The dual variables ν_j can be non-zero even if $h_j(\mathbf{x}^*) = 0$ (which it must be at feasibility).

Remark (Board Diagram Suggestion). Create a simple table showing two inequality constraints in a 2D problem:

Case	$g_i(\mathbf{x}^*)$	λ_i^*
Active constraint	$= 0$	Could be > 0
Inactive constraint	< 0	Must be 0

This table crystallizes the complementary slackness concept.

5 Theorem 4: Duality of the Fractional Knapsack Problem

We now apply duality theory to the fractional knapsack problem, which provides concrete intuition for strong duality and complementary slackness.

5.1 Problem Setup

Definition 5 (Fractional Knapsack LP). *The fractional knapsack problem is:*

$$\max_x c^T x \quad (4)$$

$$\text{s.t. } a^T x \leq b \quad (5)$$

$$0 \leq x \leq 1 \quad (6)$$

where $c, a \in \mathbb{R}^n$ with $a, c > 0$ (element-wise), $b > 0$, and we interpret x_i as the fraction of item i to include in the knapsack.

5.2 Deriving the Dual

To write the dual, we first put the problem in standard form with inequality constraints:

$$\max_x c^T x \quad (7)$$

$$\text{s.t. } a^T x \leq b \quad (8)$$

$$-x \leq 0 \quad (9)$$

$$x \leq 1 \quad (10)$$

Converting to minimization (multiply the objective by -1):

$$\min_x (-c)^T x \quad (11)$$

$$\text{s.t. } a^T x \leq b \quad (12)$$

$$-x \leq 0 \quad (13)$$

$$x - \mathbf{1} \leq 0 \quad (14)$$

The Lagrangian is:

$$\mathcal{L}(x, \lambda_0, \lambda, \mu) = -c^T x + \lambda_0(a^T x - b) + \lambda^T(-x) + \mu^T(x - \mathbf{1})$$

where $\lambda_0, \lambda, \mu \geq 0$.

Rearranging:

$$\mathcal{L}(x, \lambda_0, \lambda, \mu) = x^T(-c + \lambda_0 a - \lambda + \mu) - \lambda_0 b - \mu^T \mathbf{1}$$

The dual function is:

$$g(\lambda_0, \lambda, \mu) = \inf_x \mathcal{L}(x, \lambda_0, \lambda, \mu)$$

For this to be bounded, we need:

$$-c + \lambda_0 a - \lambda + \mu = 0 \quad \Rightarrow \quad \lambda_0 a_i - \lambda_i + \mu_i = c_i$$

Under this condition:

$$g(\lambda_0, \lambda, \mu) = -\lambda_0 b - \mu^T \mathbf{1}$$

The dual problem is:

$$\max_{\lambda_0, \lambda, \mu} \quad -\lambda_0 b - \sum_{i=1}^n \mu_i \quad (15)$$

$$\text{s.t.} \quad \lambda_0 a_i - \lambda_i + \mu_i = c_i, \quad i = 1, \dots, n \quad (16)$$

$$\lambda_0, \lambda, \mu \geq 0 \quad (17)$$

Equivalently (converting back to maximization):

$$\min_{\lambda_0, \lambda, \mu} \quad \lambda_0 b + \sum_{i=1}^n \mu_i \quad (18)$$

$$\text{s.t.} \quad \lambda_0 a_i + \mu_i \geq c_i, \quad i = 1, \dots, n \quad (19)$$

$$\lambda_0, \mu \geq 0 \quad (20)$$

5.3 The Candidate Optimal Dual Solution

To find the optimal dual solution, consider the fractional knapsack algorithm: order items by *efficiency* c_i/a_i in decreasing order, and greedily fill the knapsack.

Let k be the critical item where the knapsack becomes full. The candidate optimal dual solution is:

$$\lambda_0^* = \frac{c_k}{a_k}, \quad \mu_i^* = \begin{cases} c_i - a_i \lambda_0^* & \text{if } i < k \\ 0 & \text{if } i \geq k \end{cases}$$

5.4 Verifying Dual Feasibility

For $i < k$:

$$\lambda_0^* a_i + \mu_i^* = \frac{c_k}{a_k} a_i + c_i - a_i \frac{c_k}{a_k} = c_i$$

So the dual constraint is **tight** ($= c_i$, not $\geq c_i$). This means $a_i x_i^* = 1$ at optimality (we fully include items $i < k$).

For $i > k$:

$$\lambda_0^* a_i + \mu_i^* = \frac{c_k}{a_k} a_i > c_i$$

assuming $a_i > a_k$ or $c_i < c_k$ in the appropriate sense. Since the efficiency of item k is highest among items $\geq k$, we have $c_i/a_i \leq c_k/a_k$, so $\lambda_0^* a_i \geq c_i$. Thus $\mu_i^* = 0$ is feasible.

For $i = k$:

$$\lambda_0^* a_k + \mu_k^* = c_k + \mu_k^*$$

From the constraint $\lambda_0^* a_i + \mu_i^* = c_i$, we get $\mu_k^* = 0$. So:

$$\lambda_0^* a_k + \mu_k^* = \frac{c_k}{a_k} a_k = c_k$$

5.5 Verifying Strong Duality

Primal objective at optimality:

$$c^T x^* = \sum_{i < k} c_i + c_k \frac{b - \sum_{i < k} a_i}{a_k}$$

Dual objective at $(\lambda_0^*, \lambda^*, \mu^*)$:

$$\lambda_0^* b + \sum_{i=1}^n \mu_i^* = \frac{c_k}{a_k} b + \sum_{i < k} (c_i - a_i \frac{c_k}{a_k})$$

Expanding:

$$= \frac{c_k}{a_k} b + \sum_{i < k} c_i - \frac{c_k}{a_k} \sum_{i < k} a_i = \sum_{i < k} c_i + \frac{c_k}{a_k} \left(b - \sum_{i < k} a_i \right)$$

This equals the primal objective, confirming $d^* = p^*$.

5.6 Economic Interpretation

The dual variable $\lambda_0^* = c_k/a_k$ is the **shadow price of capacity**. It represents the marginal value of an additional unit of capacity in the knapsack. Specifically:

- If we increase b by ϵ , the optimal objective increases by approximately $\lambda_0^* \epsilon = \frac{c_k}{a_k} \epsilon$.
- This is exactly the value of adding a fractional amount of the marginal item k .

The complementary slackness conditions are:

- If $i < k$: $\mu_i^* > 0$ and $x_i^* = 1$ (we include the item fully)
- If $i > k$: $\mu_i^* = 0$ and $x_i^* = 0$ (we don't include the item)
- If $i = k$: $0 \leq x_k^* \leq 1$ (we include a fraction)

5.7 Teaching Notes

Key Insight. Why Fractional Knapsack is Special: Fractional knapsack is one of the few discrete/combinatorial problems where a greedy algorithm is provably optimal. The duality theory we've developed explains why: the greedy solution satisfies the KKT conditions (which include strong duality and complementary slackness), making it optimal.

Remark (Pedagogical Value). 1. **Intuition for shadow prices:** The fraction c_k/a_k has a clear interpretation. Students see that the “bang per buck” (value per weight) determines how much each unit of capacity is worth.

2. **Verification of optimality:** By computing the dual and verifying strong duality, students see a concrete example of duality in action. They can check the fractional knapsack solution against the dual to confirm optimality.

3. **Greedy optimality:** Many students find it surprising that the greedy algorithm is optimal. Duality provides the theoretical justification.

Remark (Common Questions). • **“Why is this convex if it's a knapsack problem?”** Because we allow fractional amounts. The 0-1 knapsack is combinatorial and non-convex; fractional knapsack is a linear program, hence convex.

- **“Does the ordering always hold?”** Yes, by definition of the efficiency metric. If items are ordered by decreasing c_i/a_i , then $c_k/a_k \geq c_i/a_i$ for all $i > k$.

- “**What if two items have the same efficiency?**” The order doesn’t matter; both solutions are optimal. Duality still holds.

Remark (Board Diagram Suggestion). Draw a simple example with 3 items:

Item	c_i	a_i	c_i/a_i
1	5	2	2.5
2	6	3	2.0
3	4	2	2.0

With $b = 5$, the greedy algorithm fills: all of item 1 (weight 2), half of item 2 (weight 3, total 5). Shadow price $= \lambda_0^* = 2.5$ (efficiency of item 1, the highest). The marginal value of capacity is 2.5 per unit.

Draw the constraint region in the feasible set to show how the dual solution emerges.

6 Summary and Connections

We have developed four key results:

1. **Weak Duality:** $d^* \leq p^*$ always holds (model-independent).
2. **Strong Duality:** Under convexity and Slater's condition, $d^* = p^*$.
3. **Complementary Slackness:** When strong duality holds, slack constraints have zero dual multipliers.
4. **Fractional Knapsack:** Demonstrates all three results in a concrete, interpretable problem.

These results form the foundation of optimization theory and are used extensively in:

- Algorithm design and analysis (Lagrangian relaxation, cutting planes)
- Economic interpretation (shadow prices, marginal values)
- Optimality conditions (KKT conditions)
- Problem decomposition (Dantzig-Wolfe, Benders decomposition)

Remark (Final Teaching Note). *Emphasize to students that duality is not just a mathematical curiosity. It is a lens through which we understand optimization problems. Every time you solve an optimization problem and check whether strong duality holds, you are (implicitly) checking whether your solution is in some sense “complete.” The dual variables tell you which constraints matter and how sensitive the solution is to changes in data. This is invaluable in practice.*